

TECH BLOG DATABASE ARCHITECTURE

A comprehensive, relational database schema optimized for modern, developer-oriented publishing platforms. Designed for structural integrity, efficient querying, and scalability.

Core Architecture Overview

This schema is designed with a traditional relational structure, strictly typed columns, and explicit foreign key constraints to ensure data integrity. It includes advanced features specific to technical blogs, such as content series clustering, markdown support flags, read-time calculations, and bookmarking mechanics.

1. users

Stores administrative authors, contributors, and registered readers.

COLUMN NAME	DATA TYPE	ATTRIBUTES	DESCRIPTION
id	BIGINT	PK, AI, UNSIGNED	Primary identifier
name	VARCHAR(255)	NOT NULL	Display name
email	VARCHAR(255)	UNIQUE, NOT NULL	Authentication & contact
password	VARCHAR(255)	NOT NULL	Hashed credential
role	ENUM	DEFAULT 'reader'	'admin', 'author', or 'reader'
github_url	VARCHAR(255)	NULLABLE	Crucial for tech authors
bio	TEXT	NULLABLE	Author biography
created_at	TIMESTAMP	NULLABLE	Account creation date
updated_at	TIMESTAMP	NULLABLE	Last update timestamp

2. series

Groups related technical posts together (e.g., "Building a POS System Part 1-5").

COLUMN NAME	DATA TYPE	ATTRIBUTES	DESCRIPTION
id	BIGINT	PK, AI, UNSIGNED	Primary identifier
title	VARCHAR(255)	NOT NULL	Series title
slug	VARCHAR(255)	UNIQUE, NOT NULL	URL friendly identifier
description	TEXT	NULLABLE	Overview of the series
created_at	TIMESTAMP	NULLABLE	Creation date
updated_at	TIMESTAMP	NULLABLE	Last update

3. posts

The central content entity. Includes metadata for SEO and performance.

COLUMN NAME	DATA TYPE	ATTRIBUTES	DESCRIPTION
id	BIGINT	PK, AI, UNSIGNED	Primary identifier
user_id	BIGINT	FK, INDEX	References users.id
category_id	BIGINT	FK, INDEX	References categories.id
series_id	BIGINT	FK, NULLABLE	References series.id
title	VARCHAR(255)	NOT NULL	Article title
slug	VARCHAR(255)	UNIQUE, NOT NULL	URL identifier
excerpt	TEXT	NULLABLE	Short summary for feeds
content	LONGTEXT	NOT NULL	Main body (Markdown/HTML)
is_markdown	BOOLEAN	DEFAULT 1	Flag for renderer
reading_time	INT	DEFAULT 0	Calculated minutes to read
status	ENUM	DEFAULT 'draft'	'draft', 'published', 'archived'
published_at	TIMESTAMP	NULLABLE, INDEX	Scheduled/Actual publish time
deleted_at	TIMESTAMP	NULLABLE	For Soft Deletes mechanism

4. categories & tags

Taxonomies for organizing content. Categories are structural, tags are descriptive.

TABLE: CATEGORIES

id	BIGINT	PK, AI	Primary identifier
name	VARCHAR(255)	NOT NULL	e.g., 'Backend', 'DevOps'
slug	VARCHAR(255)	UNIQUE	URL identifier

TABLE: TAGS

id	BIGINT	PK, AI	Primary identifier
name	VARCHAR(255)	NOT NULL	e.g., 'Laravel', 'Docker'
slug	VARCHAR(255)	UNIQUE	URL identifier

5. post_tag (Pivot Table)

Resolves the many-to-many relationship between posts and tags.

COLUMN NAME	DATA TYPE	ATTRIBUTES	DESCRIPTION
post_id	BIGINT	FK, INDEX	References posts.id (Cascade Delete)
tag_id	BIGINT	FK, INDEX	References tags.id (Cascade Delete)

6. comments

Allows for nested community discussions.

COLUMN NAME	DATA TYPE	ATTRIBUTES	DESCRIPTION
id	BIGINT	PK, AI	Primary identifier
post_id	BIGINT	FK, INDEX	References posts.id
user_id	BIGINT	FK, INDEX	References users.id
parent_id	BIGINT	FK, NULLABLE	References comments.id (Self-referential)
content	TEXT	NOT NULL	Comment body
is_approved	BOOLEAN	DEFAULT 1	For moderation purposes
created_at	TIMESTAMP	NULLABLE	Date submitted

7. bookmarks (Interactions)

Allows logged-in readers to save technical tutorials for offline or later reading.

COLUMN NAME	DATA TYPE	ATTRIBUTES	DESCRIPTION
id	BIGINT	PK, AI	Primary identifier
user_id	BIGINT	FK, INDEX	References users.id (Cascade Delete)
post_id	BIGINT	FK, INDEX	References posts.id (Cascade Delete)
created_at	TIMESTAMP	NULLABLE	Date bookmarked